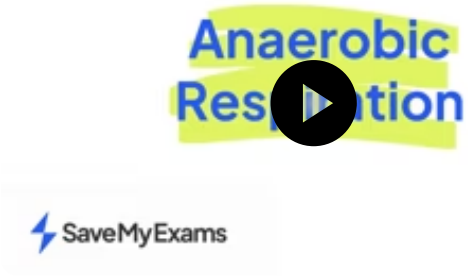
Anaerobic Respiration

Exam code: 0610 & 0970

Download PDF

Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
Updated on 25 September 2024



Did this video help you? Yes No

Anaerobic Respiration – Respiration Without Oxygen

Help me understand

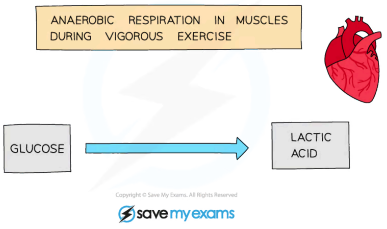
- **Anaerobic** respiration does **not require oxygen** and is defined as **the chemical reactions in cells that break down nutrient molecules to release energy without using oxygen**
- It is the **incomplete breakdown of glucose** and releases a **relatively small amount** of energy (compared to aerobic respiration) for use in cell processes
- It produces different breakdown products depending on the type of organism it is taking place in
- You need to know the equations for anaerobic respiration in **humans** (animals) and the microorganism **yeast**

Anaerobic Respiration in Animals

- Anaerobic respiration mainly takes place in muscle cells during vigorous exercise
- When we exercise vigorously, our muscles have a **higher demand for energy** than when we are resting or exercising normally. Our bodies can only deliver so much oxygen to our muscle cells for aerobic respiration
- In this instance, as much glucose as possible is broken down with **oxygen**, and some glucose is broken down without it, producing **lactic acid** instead

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

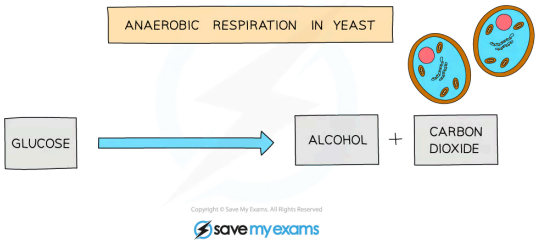
anaerobically



Word equation for anaerobic respiration in animals

Anaerobic Respiration in Yeast

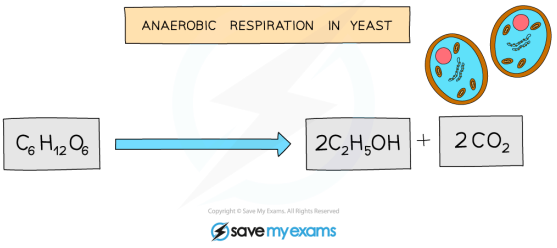
- We take advantage of the products of anaerobic respiration in **yeast** by using it in **bread making**, where
  - The carbon dioxide produced causes dough to rise
- And in **brewing**, where
  - The ethanol produced gives the beer its alcoholic nature
  - The carbon dioxide produced gives beer its fizz



Word equation for anaerobic respiration in yeast

Balanced Chemical Equation for Anaerobic Respiration: Extended

- The balanced chemical equation for anaerobic respiration in yeast is:



Balanced equation for anaerobic respiration in yeast

Oxygen Debt: Extended

- **Lactic acid** builds up in muscle cells and **lowers the pH** of the cells (making them more acidic)
- This could **denature the enzymes in cells** so it needs to be removed
- Cells excrete lactic acid into the blood. When blood passes through the **liver**, lactic acid is taken up into liver cells where it is **oxidised,** producing carbon dioxide and water (Lactic acid reacts with oxygen - this is actually **aerobic respiration** with lactic acid as the nutrient molecule instead of glucose)
- So the waste products of lactic acid oxidation are carbon dioxide and water
- This is the reason we **continue to breath heavily** and our **heart rate remains high** even after finishing exercise - we need to transport the lactic acid from our muscles to the liver, and continue getting larger amounts of oxygen into the blood to oxidise the lactic acid
- This is known as '**repaying the oxygen debt**'

Examiner Tips and Tricks

It's easy to get confused about the products of anaerobic respiration in animals: The ONLY product made is **lactic acid**. Carbon dioxide is NOT one of the products made in anaerobic respiration in animals: It is made in **aerobic** respiration!

Comparing Aerobic & Anaerobic Respiration

Comparing types of respiration:

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Test yourself Flashcards

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.